Online Appendix — Votes Without Neighborhoods

This appendix gives the formal framework and proofs (A), the data (B), the results cited in the text (C) and the re-implementation of a published allocation (D). Further tables, and the code that produces every number, are in the replication files.

A. Formal framework, proofs and computation

Parameters and symbols. η: uniform movement cap (share of a station's ballots cast by non-residents beyond the movement the roll forces); λ: multiple of the mobility-calibrated cap; δ: tolerance for residents' out-of-zone composition around their station; s: tolerance for residents' early composition beyond their station's own departure from the benchmark a^F; t: tolerance for movers' composition around their home station; κ: bound on the within-jurisdiction slope of residents' early-vote departures on log price (Section 4.5). s^*, λ^*, η^*, δ^*, κ^*: the largest value of each parameter at which the lower bound of a slope stays positive. z: roll-based share; y: valid-vote share. Columns A–F: the assumption sets of Table 2.

A.1 Observables, unknowns and the polytope

Consider one counting jurisdiction g: a district (gu), or a National Assembly constituency in 2024 (48 units). It contains neighborhoods (dongs) i and in-zone early stations j, one per dong. Ballot categories c are candidates and invalid ballots. The returns and the roll give, for each dong, election-day ballots A_{ic} (cast by residents of i), in-zone early ballots K_{jc} counted at station j, the certified electorate E_i and the electorate remaining on the election-day roll D_i, which is printed after early voting with early voters marked and excludes residents registered for at-home, shipboard or overseas voting. $E_i - D_i$ is the number of residents removed from the election-day roll. Jurisdiction totals are the out-of-zone early ballots P_c, the at-home, shipboard and overseas ballots Q_c, and $U = \sum_i (E_i - D_i) - \sum_j K_j - \sum_c P_c - \sum_c Q_c$, the removed residents who cast no counted ballot. Write $A_i = \sum_c A_{ic}$, $K_j = \sum_c K_{jc}$, $H_i = A_i + E_i - D_i$ and $r_j = \max\{0, K_j - (E_j - D_j)\}$.

The unknowns are F_{ijc} (in-zone early ballots of residents of i cast at j), P_{ic} and Q_{ic} (residents' pooled ballots) and U_i (removed residents who did not vote), all non-negative and subject to

$$\sum_{j,c} F_{ijc} + \sum_c P_{ic} + \sum_c Q_{ic} + U_i = E_i - D_i, \tag{R}$$

$$\sum_i F_{ijc} = K_{jc}; \quad \sum_i P_{ic} = P_c; \quad \sum_i Q_{ic} = Q_c; \quad \sum_i U_i = U, \tag{S}$$

$$F_{ijc} = 0 \ \text{ if } i \text{ and } j \text{ lie in different early-voting zones.} \tag{Z}$$

Residents' count of category c is $N_{ic} = A_{ic} + \sum_j F_{ijc} + P_{ic} + Q_{ic}$. The roll-based share is $z_{ic} = N_{ic}/H_i$ and the valid-vote share $y_{ic} = N_{ic}/(H_i - U_i - N_{i,\text{inv}})$.

Restrictions. (M_η) caps non-resident ballots at station j at $r_j + \eta K_j$; the mobility-calibrated scenario caps them at $r_j^\rho + \lambda C_j$, with $r_j^\rho = \max\{r_j, K_j - \phi_u(E_j - D_j)\}$, ϕ_u the zone's ratio of in-zone early ballots to removed residents, and C_j the calibrated cross-dong flow (B.4). (A_δ) bounds residents' out-of-zone composition around an anchor $\tilde{a}_{ic}$: $(\tilde{a}_{ic} - \delta)P_{i\cdot} \le P_{ic} \le (\tilde{a}_{ic} + \delta)P_{i\cdot}$, with $\tilde{a}_{ic} = K_{ic}/K_i$ (the station's composition; in 2012, without in-zone early voting, the election-day composition), or the station composition shifted by the jurisdiction's out-of-zone minus in-zone early composition, optionally plus its price relation. (A_s^F) bounds residents' in-zone early composition around $a_{ic}^F = \mathrm{clip}_{[0,1]}(A_{ic}/A_i + \gamma_{gc}^K)$, with γ_{gc}^K the jurisdiction's in-zone early minus election-day composition:

$$(a_{ic}^F - \omega_{ic}) \sum_{j,c'} F_{ijc'} \le \sum_j F_{ijc} \le (a_{ic}^F + \omega_{ic}) \sum_{j,c'} F_{ijc'}, \qquad \omega_{ic} = |K_{ic}/K_i - a_{ic}^F| + s,$$

with the lower limit truncated at zero. (A_l^{mv}) bounds movers: with $k_{ic} = K_{ic}/K_i$, $(k_{ic} - t) \sum_{j\neq i,c'} F_{ijc'} \le \sum_{j\neq i} F_{ijc} \le (k_{ic} + t) \sum_{j\neq i,c'} F_{ijc'}$. ($A^\kappa$) bounds the alignment of residents' early-vote departures with price. Let $\Delta_i = \sum_j F_{ij,\mathrm{cons}} - a_{i,\mathrm{cons}}^F \sum_{j,c} F_{ijc} + P_{i,\mathrm{cons}} - \tilde{a}_{i,\mathrm{cons}} P_{i\cdot}$, let I_u be the dongs of counting jurisdiction u with a price (in 2024 a constituency, not a gu), $\bar{x}_u$ their mean log price, and $v_i = (x_i - \bar{x}_u) / \sum_{j\in I_u} (x_j - \bar{x}_u)^2$ for $i \in I_u$ and 0 otherwise, so that $\sum_{i\in u} v_i = 0$. Then $|\sum_{i\in u} v_i \Delta_i / H_i| \le \kappa$ for every u: the within-jurisdiction OLS slope of departures, as shares of the roll, on log price. (A_P^κ) is the same restriction on out-of-zone departures alone, $\Delta_i^P = P_{i,\mathrm{cons}} - \tilde{a}_{i,\mathrm{cons}} P_{i\cdot}$, used in place of ($A_\delta$). The observed reference for κ applies the same weights to station departures $(K_{i,\mathrm{cons}} - a_{i,\mathrm{cons}}^F K_i)/H_i$. Λ is the set of non-negative unknowns satisfying (R), (S), (Z) and the imposed restrictions.

A.2 Results

Lemma 1. Under (R), (S) and non-negativity, $\sum_{i\neq j} \sum_c F_{ijc} \ge r_j$ for every j.

Proof. By (S), $\sum_{i\neq j} \sum_c F_{ijc} = K_j - \sum_c F_{jjc}$, and by (R) $\sum_c F_{jjc} \le E_j - D_j$. □

Corollary 1. Under (M_0), every dong j with $r_j > 0$ has $P_{j\cdot} = Q_{j\cdot} = U_j = 0$ and $F_{jk\cdot} = 0$ for $k \neq j$: its removed residents all voted at their own station.

Proof. Non-resident ballots at j are at most r_j, so $F_{jj\cdot} \ge K_j - r_j = E_j - D_j$, and (R) forces every other term to zero. □

Summing (R) over the dongs of a zone u and using (S) and (Z) gives a zone-level identity, $\sum_{i\in u}(P_{i\cdot} + Q_{i\cdot} + U_i) = \sum_{i\in u}(E_i - D_i) - \sum_{j\in u} K_j$.

Proposition 1. Λ is a convex polytope. For any weights w, the set of values of $\sum_i w_i z_{ic}$ over Λ is $[\min_\Lambda, \max_\Lambda]$, each endpoint the value of a linear program.

Proof. All constraints are linear in the unknowns, and z_{ic} is linear because H_i is observed; a linear function on a convex polytope attains every value between its extremes. □

The bounds are sharp for the continuous relaxation, relative to the information in (R), (S), (Z) and the restrictions. Jurisdictions share no constraint, so citywide extrema are sums of jurisdiction-specific extrema.

Proposition 2. Let $w = w^{\mathrm{btw}} + w^{\mathrm{wth}}$ with $w_i^{\mathrm{btw}} = H_i \mu_{g(i)}$. (i) $\sum_i w_i^{\mathrm{btw}} z_{ic}$ is point-identified by the jurisdiction totals. (ii) The identified set of $\sum_i w_i z_{ic}$ is that number plus the identified set of $\sum_i w_i^{\mathrm{wth}} z_{ic}$. (iii) Without ($A_\delta$), ($A^\kappa$) and ($A_P^\kappa$), if some point of Λ has $P_{ic'} > 0$ and $P_{i'c} > 0$ for dongs $i \neq i'$ of one jurisdiction, the slope is not point-identified whenever $w_i/H_i \neq w_{i'}/H_{i'}$. (iv) The width is at most $\sum_g (\max_{i\in g} w_i/H_i - \min_{i\in g} w_i/H_i) M_{gc}$, with $M_{gc} = P_{gc} + Q_{gc} + \sum_{j\in g} \min\{K_{jc}, m_j\}$ and m_j the movement cap.

Proof. (i) $\sum_{i\in g} N_{ic}$ equals the observed jurisdiction total. (ii) Linearity in w. (iii) Exchanging ε ballots between $(i,c),(i,c'),(i',c),(i',c')$ leaves every constraint unchanged and moves the slope by $\varepsilon(w_i/H_i - w_{i'}/H_{i'})$. (iv) Between two points of Λ, $\sum_{i\in g}(N_{ic} - N'_{ic}) = 0$ and the positive parts are bounded by M_{gc}, because both points keep at least $K_{jc} - m_j$ of station j's category-c ballots with its residents; hence $\sum_{i\in g}(w_i/H_i)(N_{ic} - N'_{ic}) \le \text{range}_g(w/H)\, M_{gc}$. □

Lemma 2. With $\pi_w = \sum_i (x_i - \bar{x}_{g(i)})^2 / \sum_i (x_i - \bar{x})^2$, the pooled OLS weights satisfy $w^P = \pi_w w^{FE} + v$, $v_i = (\bar{x}_{g(i)} - \bar{x}) / \sum_k (x_k - \bar{x})^2$, so for every $\ell \in \Lambda$, $\theta(w^P;\ell) = \pi_w \theta(w^{FE};\ell) + \theta(v;\ell)$.

Proof. Decompose $x_i - \bar{x}$ into within- and between-gu parts and use linearity. □

$\theta(v;\ell)$ is a gradient computed from residence-based dong shares; it is not the gradient of gu totals and depends on the allocation. It distinguishes the between-district component of a dong-level regression from a regression on fully counted district totals.

Proposition 3. Λ weakly expands as we increase η, λ, δ, s and t, and when any restriction, including (Z), is removed; so the lower bound of a z slope is non-increasing in each parameter and breakdown points are well defined.

Proof. Each change relaxes inequalities or removes equalities. □

Proposition 4. With $T_i = H_i - U_i - N_{i,\text{inv}}$, $y_{ic} - z_{ic} = y_{ic}(U_i + N_{i,\text{inv}})/H_i$, so with dong bounds $\underline{y}_{ic} \le y_{ic} \le \bar{y}_{ic}$ over Λ, $w_i y_{ic} \ge w_i z_{ic} + k_i(U_i + N_{i,\text{inv}})/H_i$ with $k_i = w_i \underline{y}_{ic}$ $(w_i > 0)$ or $w_i \bar{y}_{ic}$ $(w_i < 0)$. The right side is linear, so one LP gives a valid lower bound on the valid-vote slope; the upper bound is symmetric.

Proof. $N/T - N/H = N(H-T)/(TH)$ and $(U_i + N_{i,\text{inv}})/H_i \ge 0$. □

A.3 Further quantities

The neighborhood contrast weights each dong priced above its gu's median by $\nu_g / n_g^{\text{high}}$ and every other dong with a price by $-\nu_g / n_g^{\text{low}}$, with ν_g proportional to the gu's number of dongs with a price.

The region of assumptions. $s^*(\lambda,\delta) = \sup\{s : \min_\Lambda \theta > 0\}$, found by bisection on $[s_{\min}, 0.30]$ (width $0.3/256$), where $s_{\min}$ is the smallest feasible slack; reported values are the lower end of the final interval.

The first reversing allocation along s attains the minimum slope at s just above s^*, holding λ and δ fixed. *The least-departure reversal* minimizes $\sum_i |\sum_j F_{ij,\text{cons}} - a^F_{i,\text{cons}} \sum_{j,c} F_{ijc}| + \sum_i |P_{i,\text{cons}} - \tilde{a}_{i,\text{cons}} P_{i\cdot}|$ subject to a non-positive slope, one LP over all jurisdictions coupled by the slope row. Each definition uses a different criterion for the smallest reversal; the second can concentrate departures in neighborhoods with the largest regression weights.

Guaranteed value of a report $S\ell$: $W_S^{\max} = \max\{c^\top(\ell - \ell') : \ell, \ell' \in \Lambda,\ S\ell = S\ell'\}$. For any true allocation ℓ^0, every ℓ with $S\ell = S\ell^0$ forms a feasible pair with it, so the range left after the report is at most $W_S^{\max}$ for every ℓ^0 and equals it for some. It separates into one LP per jurisdiction. *Revealing the direction.* Report S reveals the direction of the slope on Λ iff no two allocations $\ell, \ell' \in \Lambda$ with $S\ell = S\ell'$ have slopes of strictly opposite sign, $\theta(\ell) < 0 < \theta(\ell')$. Allocations with $\theta = 0$ count as neither sign, so full disclosure ($S\ell = S\ell' \Rightarrow \ell = \ell'$) always reveals the direction. Let $t_S^* = \max\{t : \theta(\ell) \le -t,\ \theta(\ell') \ge t,\ S\ell = S\ell',\ \ell, \ell' \in \Lambda\}$, one LP over all jurisdictions. Because the set of pairs is symmetric in (ℓ, ℓ'), a strictly opposite pair exists iff $t_S^* > 0$, and t_S^* is then the largest margin of such a pair. Hence S reveals the direction iff $t_S^* \le 0$. Table C16 reports t_S^*. Under every assumption set and report we consider, $t_S^* > 0$ wherever the current set contains zero; in 2022 under the main assumptions $t_S^* = 0.026$ for residence reporting.

A.4 Computation

Each jurisdiction is one LP with $n^2|\mathscr{C}| + 2n|\mathscr{C}| + n$ variables ($|\mathscr{C}| = 6$). All restrictions are built by one routine used by every computation, including the valid-vote programs and the integer checks. Programs are solved with HiGHS through SciPy, with objective coefficients normalized to unit maximum (without it, coefficients of order 10^{-9} fall below the solver's dual tolerance and bounds are reported too narrow) and feasibility tolerances of 10^{-9} and 10^{-10}. Exact dong-level valid-vote bounds use Dinkelbach's (1967) algorithm. Ballots are integers: Table A1 re-solves near-zero lower bounds as mixed-integer programs with a 30-second limit per solve.

Table A1. Integer check of near-zero lower bounds

Cell	LP z lower	Integer z lower (solver bound)	Integer z incumbent	Prop. 4 y lower (integer, solver bound)	y at integer allocation	Solves at time limit
2012 within-district, $\eta = 0$, $\delta = 1$	0.0077	0.0077	0.0077	0.0069	0.0070	0/50
2026 within-district, $\eta = 0.1$, $\delta = 0.1$	0.0137	0.0139	0.0139	0.0098	0.0106	10/50
2026 within-district, calibrated, $\delta = 0.1$	0.0104	0.0106	0.0106	0.0065	0.0072	7/50
2025 within-district, $\eta = 0$, $\delta = 0.1$	0.0017	0.0018	0.0018	−0.0020	−0.0006	2/50
2024 within-constituency, $\eta = 0$, $\delta = 0.1$	0.0036	0.0037	0.0037	0.0078	0.0092	1/96
2022 within-district, $\eta = 0$, $\delta = 0.1$	0.0144	0.0145	0.0145	0.0187	0.0204	3/50

Notes. Mixed-integer programs use the same constraint matrices as the LPs, including the 2026 zones. Each solve has a 30-second limit, extended to 180 seconds when no integer solution has been found. Solver bound: the best bound proven by the solver when it stops, which is a valid lower bound on the integer minimum also when the time limit is reached; the incumbent is the best integer allocation found.

B. Data

B.1 Returns and rolls. Returns are the National Election Commission's counting-unit files (2012–2025) and counting-unit spreadsheets (2026) (National Election Commission 2026). Each dong contributes its election-day voting districts and its in-zone early station; each gu, or constituency in 2024, its out-of-zone early, at-home (and shipboard) and overseas rows. Certified electorates by dong come from the NEC electorate service; in presidential and mayoral elections its totals equal the counting files' totals in all 25 gu. In presidential elections overseas electors with resident registration are assigned to dongs in E_i and are not on the election-day roll. In 2024 the service exceeds the constituency files by 11,672 electors (0.14%), overseas electors without resident registration who vote only for the party list; they are added to U. U is non-negative in every unit and differs from the published abstentions of the pooled modes by −2 to 57 per gu; positive differences are in-zone early abstentions, and the small negative differences reflect minor inconsistencies between the certified electorate and the counting files. U is small, 0.33%, 0.40%, 0.25% and 0.03% of H in 2022, 2024, 2025 and 2026 (Table C25), and larger in higher-price districts, where more residents are registered abroad; it is allocated among a jurisdiction's dongs without restriction.

In 2022, voters with COVID-19 voted early in a separate session on the second early-voting day or on election day after regular hours; the counting-unit files contain no separate category for these ballots, which are reported within the standard modes and treated accordingly.

B.2 The 2026 zones. When local councils and heads are elected together at the end of their terms, the early-voting zone is the district-council constituency (Regulations on Public Official Elections, Art. 133(3); National Election Commission n.d.); in presidential elections it is the gu, and in National Assembly elections the constituency where a gu holds two or more (Art. 86(3)). The NEC's table of council constituencies and dongs (*Table of Constituencies and Eup, Myeon and Dong*, 선거구및읍면동현황) assigns each of

Seoul's 427 dongs to one of 155 constituencies. For the 113 contested constituencies, the council returns list the same dongs and report out-of-zone early and at-home voters. A voter receives all ballots at once, so out-of-zone status is the same in every contest: wherever all of a gu's constituencies were contested, the council counts sum exactly to the mayoral gu count, and at each of the 320 in-zone early stations of contested constituencies the council and mayoral counts of early voters are equal (Table B1).

Table B1. 2026 early-voting zones: district-council constituencies

Gu	Dongs	Council constituencies	Contested	Out-of-zone early voters in contested constituencies (council returns)	Out-of-zone early voters, gu (mayoral returns)	Sum check where all contested
Gangnam-gu	22	9	7	26,033	33,020	–
Gangdong-gu	19	7	6	26,091	30,597	–
Gangbuk-gu	13	4	3	13,577	17,776	–
Gangseo-gu	20	8	8	36,021	36,021	exact
Gwanak-gu	21	7	7	32,557	32,557	exact
Gwangjin-gu	15	4	4	19,980	19,980	exact
Guro-gu	16	6	4	16,256	23,100	–
Geumcheon-gu	10	4	4	15,188	15,188	exact
Nowon-gu	19	6	4	20,751	32,117	–
Dobong-gu	14	5	4	16,008	19,875	–
Dongdaemun-gu	15	6	6	24,547	24,547	exact
Dongjak-gu	15	7	5	24,860	31,822	–
Mapo-gu	16	8	5	18,519	28,812	–
Seodaemun-gu	14	5	4	17,119	23,020	–
Seocho-gu	18	5	5	23,753	23,753	exact
Seongdong-gu	17	4	3	13,466	17,606	–
Seongbuk-gu	20	7	3	15,460	30,800	–
Songpa-gu	27	10	6	25,683	41,101	–
Yangcheon-gu	18	8	4	11,429	28,888	–
Yeongdeungpo-gu	18	7	3	12,670	29,215	–
Yongsan-gu	16	5	3	8,913	14,431	–
Eunpyeong-gu	16	8	5	21,224	33,789	–
Jongno-gu	17	4	2	4,098	8,858	–
Jung-gu	15	4	3	6,290	9,197	–
Jungnang-gu	16	7	5	20,046	25,358	–

Notes. Source: NEC Election Statistics System, constituency-and-dong table and constituency-level council returns. In every gu where all constituencies were contested, the council counts sum exactly to the mayoral gu count. In-zone early voters per station: council and mayoral counts are equal at 320 of 320 stations of contested constituencies.

B.3 Prices. Apartment transactions in the 12 months before each election (Ministry of Land, Infrastructure and Transport 2026) are geocoded from lot number to administrative dong through the road-name address service (91% to over 99% matched exactly; the rest by same-lot or nearest-lot majority within the legal dong, or dropped) and mapped to the dong names in force at each election (Yongdu-dong and Sinseol-dong to Yongsin-dong; Gaepo 3-dong to Irwon 2-dong before 2023; the five Sindang dongs to their pre-2014 names in 2012; Sangil 1- and 2-dong to Sangil-dong and Hang-dong to Oryu 2-dong in 2017). The regressor is the log median price per square meter. In 2017–2026 the median dong has 70–200 transactions; dongs with fewer than 30 number 99, 38, 96, 109, 62 and 43 in 2012, 2017, 2022, 2024, 2025 and 2026.

B.4 Mobility. Seoul Living Mobility (Seoul Metropolitan Government and KT 2026) reports trips by origin and destination dong, day and hour, with a stay of at least 27 minutes; counts of three or fewer are suppressed. We keep home-origin trips from Seoul arriving 06:00–17:59 on the two weekdays of early voting (Friday and Saturday; Thursday and Friday in 2025), summed over the month's days with those weekdays; a calibration from each day alone gives nearly the same bounds (replication files, calibrated_dow). For zone u, W_u counts trips to another dong of u and X_u trips leaving u; $R_u = W_u/X_u$ and $C_u = R_u B_u$, with B_u the zone's out-of-zone early ballots (in 2026 the council headcount for contested constituencies, scaled to counted mayoral ballots, and otherwise the gu residual split in proportion to removed residents who did not vote at an in-zone station, $E - D - K$); C_u is allocated to stations by their inflow from other dongs of u. Suppressed counts are set to 2. The mobility data use an older dong system in a few places, where one code covers two election dongs (Sangil 1- and 2-dong; Oryu 2-dong and Hang-dong; Sinseol-dong and Yongdu-dong); trips of such a code are split between its dongs in proportion to their electorates, and in the one 2026 zone that consists only of two such dongs, where movement between them is unobservable, R_u is set to the median of the other zones with two or more dongs. Calibrated flows are a median 9–21% of a station's ballots in 2022–2025 and 4% in 2026.

B.5 Counting rules elsewhere.

Table B2. Counting rules for ballots cast away from the assigned polling place

Country	Smallest unit with candidate totals	Ballots reported above that unit	Usual treatment
Taiwan	Polling station, village	None in general	Read directly; residence-based by household registration
Japan	Counting district (開票区)	All ballots mixed at the counting district	Not reported below the counting district
United States (many states)	Precinct	Absentee, early, provisional ballots (county or city)	Allocated in proportion to in-person votes
Germany	In-person precinct	Postal ballots, counted in postal districts	Analyst's choice
Canada	Polling division	Special-voting-rules ballots (electoral district)	Analyst's choice
Australia	Polling place	Declaration votes (division)	Analyst's choice
Korea	Voting district, dong	Out-of-zone early, at-home, overseas ballots (gu or constituency)	Dropped, allocated or imputed

Sources. Taiwan: Civil Servants Election and Recall Act, Art. 17 (Taiwan n.d.a, n.d.b). Japan: Public Offices Election Act, Arts. 18 and 66(2) (Japan 1950). United States: Amos et al. (2017); Palmer and Schneer (2021). Germany: Bundeswahlordnung § 7 (Germany 2002); postal ballots were 47.3% of ballots in 2021 and 37.0% in 2025 (Die Bundeswahlleiterin 2025). Canada: Canada Elections Act, s. 168(1) (Canada 2000); Elections Canada (2025, Table 5). Australia: Australian Electoral Commission (2025).

C. Results cited in the text

Table C1 gives identified sets under uniform movement for all elections, citywide and within gu, with breakdown points. Table C2 shows what the 2026 zones and the council contest's headcounts add (Section 4.4). Tables C3–C7 concern the assumptions of Section 4.5: mobility-calibrated movement, the region where the within-district slope is positive, the allocations that reverse it, and restrictions on how movers vote. Tables C8–C11 are robustness checks: the early-vote gap and price, the out-of-zone benchmark, alternative specifications and the neighborhood contrast. Table C12 reports guaranteed reporting gains, Table C13 the sensitivity to price measurement, Table C14 the regressions across districts, Ta-

ble C15 valid-vote shares, Table C16 whether a report reveals the direction of the slope, Table C17 the analysis without Gangnam, Seocho and Songpa (Section 4.3) for columns A–E of Table 2, Table C18 the same for valid-vote shares, Table C19 how far the residents' early-composition assumption can be relaxed with and without the three districts, Table C20 the point-identified benchmarks: election-day voters and in-zone station shares (Section 4.2), Table C21 the observed gaps between out-of-zone and in-zone early voters that serve as a reference for the out-of-zone tolerance (Section 3.4), Table C22 the restriction on price-aligned departures, Table C23 age composition and the tercile contrast, and Table C24 alternative outcomes, including the combined conservative vote in 2017, and Table C25 further restrictions: the largest out-of-zone tolerance, movers' composition together with residents' composition, and price-aligned out-of-zone departures.

Table C1. Identified sets under uniform movement and breakdown points η^* (z; bold = contains zero)

Election	Conv.	Accounting only	$\eta = 0$, $\delta = 1$	$\eta = 0$, $\delta = \delta_{\min}$	$\eta = 0$, $\delta = 0.1$	$\eta = 0.05$, $\delta = 0.1$	$\eta = 0.1$, $\delta = 0.1$	$\eta = 0.2$, $\delta = 0.1$	η^*, $\delta = 0.1$	η^*, $\delta = 1$
Citywide										
2012	0.123	[0.087, 0.151]	[0.087, 0.151]	[0.102, 0.138]	[0.096, 0.143]	[0.096, 0.143]	[0.096, 0.143]	[0.096, 0.143]	any	any
2017	0.049	**[−0.135, 0.232]**	**[−0.015, 0.110]**	[0.036, 0.056]	[0.016, 0.077]	**[−0.015, 0.110]**	**[−0.041, 0.135]**	**[−0.078, 0.173]**	0.021	0
2022	0.150	**[−0.244, 0.484]**	[0.040, 0.240]	[0.132, 0.166]	[0.103, 0.190]	[0.064, 0.225]	[0.029, 0.258]	**[−0.035, 0.319]**	0.143	0.072
2024	0.150	**[−0.148, 0.406]**	[0.067, 0.215]	[0.135, 0.156]	[0.117, 0.175]	[0.084, 0.204]	[0.056, 0.231]	[0.004, 0.281]	0.207	0.150
2025	0.119	**[−0.114, 0.324]**	[0.040, 0.185]	[0.101, 0.136]	[0.085, 0.149]	[0.055, 0.175]	[0.030, 0.199]	**[−0.016, 0.243]**	0.162	0.096
2026	0.133	**[−0.030, 0.276]**	[0.069, 0.187]	[0.128, 0.138]	[0.119, 0.146]	[0.106, 0.159]	[0.093, 0.171]	[0.069, 0.195]	any	0.311
Within-district										
2012	0.087	[0.008, 0.157]	[0.008, 0.157]	[0.044, 0.127]	[0.027, 0.140]	[0.027, 0.140]	[0.027, 0.140]	[0.027, 0.140]	any	any
2017	0.014	**[−0.428, 0.450]**	**[−0.136, 0.162]**	**[−0.013, 0.034]**	**[−0.060, 0.085]**	**[−0.136, 0.163]**	**[−0.199, 0.225]**	**[−0.288, 0.314]**	0	0
2022	0.097	**[−0.615, 0.704]**	**[−0.100, 0.264]**	[0.068, 0.131]	[0.014, 0.175]	**[−0.057, 0.239]**	**[−0.122, 0.298]**	**[−0.238, 0.411]**	0.008	0
2024	0.102	**[−0.553, 0.683]**	**[−0.077, 0.254]**	[0.071, 0.121]	[0.033, 0.164]	**[−0.041, 0.230]**	**[−0.105, 0.290]**	**[−0.218, 0.402]**	0.020	0
2025	0.077	**[−0.468, 0.552]**	**[−0.104, 0.232]**	[0.037, 0.119]	[0.002, 0.149]	**[−0.068, 0.211]**	**[−0.126, 0.265]**	**[−0.230, 0.365]**	0	0
2026	0.103	**[−0.279, 0.439]**	**[−0.046, 0.234]**	[0.093, 0.117]	[0.072, 0.138]	[0.043, 0.167]	[0.014, 0.195]	**[−0.041, 0.250]**	0.123	0
Within-constituency										
2024	0.089	**[−0.729, 0.816]**	**[−0.134, 0.279]**	[0.051, 0.114]	[0.004, 0.166]	**[−0.088, 0.249]**	**[−0.167, 0.324]**	**[−0.310, 0.464]**	0	0

Notes. Accounting only: movement unrestricted ($\eta = 1$, at which the cap $r_j + K_j$ cannot bind) and $\delta = 1$. $\delta_{\min}$: each jurisdiction at its smallest feasible out-of-zone band. η^*: largest η at which the lower bound is positive, lower end of a bisection interval of width 1/512 on [0, 1] ("any": positive at $\eta = 1$). 2026 imposes the statutory early-voting zones.

Table C2. 2026: the statutory early-voting zones and cross-contest headcounts (z)

Cell	Slope	Movement anywhere in the gu (zones ignored)	Statutory zones: district-council constituencies (main)	Zones + council-contest headcounts
accounting only	Within-district	**[−0.378, 0.519]**	**[−0.279, 0.439]**	**[−0.279, 0.439]**
accounting only	Citywide	**[−0.071, 0.308]**	**[−0.030, 0.276]**	**[−0.030, 0.276]**
minimum movement, δ=1	Within-district	**[−0.046, 0.234]**	**[−0.046, 0.234]**	**[−0.046, 0.234]**
minimum movement, δ=1	Citywide	[0.069, 0.187]	[0.069, 0.187]	[0.069, 0.187]
minimum movement, δ=0.1	Within-district	[0.072, 0.139]	[0.072, 0.138]	[0.075, 0.137]
minimum movement, δ=0.1	Citywide	[0.118, 0.147]	[0.119, 0.146]	[0.120, 0.146]
eta=0.1, δ=0.1	Within-district	**[−0.039, 0.239]**	[0.014, 0.195]	[0.016, 0.194]
eta=0.1, δ=0.1	Citywide	[0.073, 0.189]	[0.093, 0.171]	[0.094, 0.171]
calibrated, δ=1	Within-district	**[−0.243, 0.419]**	**[−0.108, 0.296]**	**[−0.108, 0.296]**
calibrated, δ=1	Citywide	**[−0.016, 0.268]**	[0.042, 0.215]	[0.042, 0.215]
calibrated, δ=0.1	Within-district	**[−0.191, 0.373]**	[0.010, 0.201]	[0.013, 0.200]
calibrated, δ=0.1	Citywide	[0.007, 0.246]	[0.092, 0.174]	[0.092, 0.173]
calibrated, δ=0.1, s=0.02	Within-district	[0.034, 0.174]	[0.054, 0.156]	[0.057, 0.154]
calibrated, δ=0.1, s=0.02	Citywide	[0.103, 0.162]	[0.111, 0.154]	[0.112, 0.153]

Notes. Zones: an in-zone early ballot can be cast only at a station in the voter's district-council constituency (155 constituencies). Council-contest headcounts: the out-of-zone early and at-home headcounts that the district-council returns report by constituency (113 contested constituencies; the uncontested ones only as a gu residual), imposed as ranges that allow for unreturned envelopes. Calibrated rows use the zone-level calibration.

Table C3. Calibrated movement, residents' early composition unrestricted (z)

Assumption	2022	2024	2025	2026
Citywide				
δ = 0.1	**[−0.062, 0.340]**	[0.018, 0.268]	**[−0.038, 0.264]**	[0.092, 0.174]
δ = 0.2	**[−0.071, 0.349]**	[0.011, 0.276]	**[−0.046, 0.271]**	[0.079, 0.187]
δ = 1	**[−0.097, 0.371]**	**[−0.010, 0.294]**	**[−0.059, 0.283]**	[0.042, 0.215]
δ = 0.1, λ = 0.5	**[−0.012, 0.296]**	[0.046, 0.242]	**[−0.007, 0.234]**	[0.097, 0.168]
δ = 0.1, λ = 1.5	**[−0.101, 0.375]**	**[−0.006, 0.289]**	**[−0.054, 0.282]**	[0.086, 0.179]
λ*, δ = 0.1	0.39	1.37	0.41	≥ 4
λ*, δ = 1	0	0.76	0.09	≥ 4
Within-district				
δ = 0.1	**[−0.283, 0.448]**	**[−0.185, 0.372]**	**[−0.283, 0.411]**	[0.010, 0.201]
δ = 0.2	**[−0.297, 0.462]**	**[−0.202, 0.388]**	**[−0.299, 0.427]**	**[−0.019, 0.230]**
δ = 1	**[−0.342, 0.499]**	**[−0.248, 0.427]**	**[−0.335, 0.453]**	**[−0.108, 0.296]**
δ = 0.1, λ = 0.5	**[−0.193, 0.367]**	**[−0.125, 0.315]**	**[−0.211, 0.342]**	[0.024, 0.187]
δ = 0.1, λ = 1.5	**[−0.354, 0.512]**	**[−0.238, 0.419]**	**[−0.321, 0.453]**	**[−0.003, 0.214]**
λ*, δ = 0.1	0	0	0	1.40
λ*, δ = 1	0	0	0	0
Within-constituency				
δ = 0.1	–	**[−0.269, 0.427]**	–	–
δ = 0.2	–	**[−0.291, 0.448]**	–	–
δ = 1	–	**[−0.351, 0.498]**	–	–
δ = 0.1, λ = 0.5	–	**[−0.193, 0.356]**	–	–
δ = 0.1, λ = 1.5	–	**[−0.337, 0.486]**	–	–
λ*, δ = 0.1	–	0	–	–
λ*, δ = 1	–	0	–	–

Notes. Cap at station j: $r_j^{\rho} + \lambda C_j$. λ*: largest λ in [0, 4] at which the lower bound is positive, lower end of a bisection interval of width 0.01.

Table C4. Sensitivity thresholds s*(λ, δ) for a positive within-district slope

Movement	δ = 0.1	0.15	0.2	0.3	0.5	1
2022						
λ=0.5	none	none	none	none	none	none
λ=1	none	none	none	none	none	none
λ=1.5	none	none	none	none	none	none

Movement	$\delta = 0.1$	0.15	0.2	0.3	0.5	1
λ=2	none	none	none	none	none	none
λ=3	none	none	none	none	none	none
unrestricted ($\eta = 2$; any $\eta \geq 1$ leaves the cap slack)	none	none	none	none	none	none
2024						
λ=0.5	0.025	0.004	none	none	none	none
λ=1	0.021	none	none	none	none	none
λ=1.5	0.018	none	none	none	none	none
λ=2	0.015	none	none	none	none	none
λ=3	0.011	none	none	none	none	none
unrestricted ($\eta = 2$; any $\eta \geq 1$ leaves the cap slack)	none	none	none	none	none	none
2025						
λ=0.5	none	none	none	none	none	none
λ=1	none	none	none	none	none	none
λ=1.5	none	none	none	none	none	none
λ=2	none	none	none	none	none	none
λ=3	none	none	none	none	none	none
unrestricted ($\eta = 2$; any $\eta \geq 1$ leaves the cap slack)	none	none	none	none	none	none
2026						
λ=0.5	∞	∞	0.144	none	none	none
λ=1	∞	0.204	0.087	none	none	none
λ=1.5	0.271	0.136	0.071	none	none	none
λ=2	0.184	0.115	0.064	none	none	none
λ=3	0.145	0.102	0.055	none	none	none
unrestricted ($\eta = 2$; any $\eta \geq 1$ leaves the cap slack)	0.122	0.079	0.006	none	none	none

Notes. ∞: positive with residents' early composition unrestricted. none: not positive at the smallest feasible slack. The sign is identified at every (λ, δ, s) with s below the entry. Bisection on [$s_{\min}$, 0.30], width 0.3/256, lower end reported.

Table C5. First allocation with a non-positive slope as s increases ($\lambda = 1$)

Election	δ	s at edge	Tercile	Movers' share	Movers' conservative share	Stayers' conservative share	Residents' early conservative share	a^F	Station share	Median |departure| of residents' early share from a^F (this allocation)	90th pct.
2022	0.1	0.000	low	0.18	0.51	0.39	0.41	0.40	0.40	0.011	0.030
2022	0.1	0.000	mid	0.28	0.52	0.38	0.42	0.42	0.43	0.011	0.030
2022	0.1	0.000	high	0.27	0.67	0.45	0.51	0.52	0.52	0.011	0.030
2022	0.15	0.000	low	0.18	0.52	0.38	0.41	0.40	0.40	0.011	0.030

Election	δ	s at edge	Tercile	Movers' share	Movers' conservative share	Stayers' conservative share	Residents' early conservative share	a^F	Station share	Median |departure| of residents' early share from a^F (this allocation)	90th pct.
2022	0.15	0.000	mid	0.28	0.53	0.38	0.42	0.42	0.43	0.011	0.030
2022	0.15	0.000	high	0.27	0.65	0.46	0.51	0.51	0.52	0.011	0.030
2022	0.2	0.000	low	0.19	0.52	0.38	0.41	0.40	0.40	0.011	0.030
2022	0.2	0.000	mid	0.27	0.54	0.38	0.42	0.42	0.43	0.011	0.030
2022	0.2	0.000	high	0.26	0.64	0.47	0.51	0.51	0.52	0.011	0.030
2024	0.1	0.022	low	0.18	0.49	0.35	0.37	0.36	0.36	0.034	0.054
2024	0.1	0.022	mid	0.14	0.51	0.34	0.37	0.37	0.37	0.034	0.054
2024	0.1	0.022	high	0.17	0.54	0.43	0.45	0.47	0.47	0.034	0.054
2024	0.15	0.000	low	0.17	0.45	0.35	0.36	0.36	0.36	0.012	0.031
2024	0.15	0.000	mid	0.15	0.48	0.35	0.37	0.37	0.37	0.012	0.031
2024	0.15	0.000	high	0.17	0.57	0.44	0.46	0.47	0.47	0.012	0.031
2024	0.2	0.000	low	0.17	0.44	0.35	0.36	0.36	0.36	0.012	0.031
2024	0.2	0.000	mid	0.15	0.49	0.35	0.37	0.37	0.37	0.012	0.031
2024	0.2	0.000	high	0.17	0.57	0.44	0.46	0.47	0.47	0.012	0.031
2025	0.1	0.000	low	0.26	0.39	0.19	0.24	0.23	0.23	0.013	0.035
2025	0.1	0.000	mid	0.33	0.39	0.16	0.23	0.23	0.23	0.013	0.035
2025	0.1	0.000	high	0.35	0.42	0.21	0.28	0.29	0.29	0.013	0.035
2025	0.15	0.000	low	0.26	0.40	0.18	0.24	0.23	0.23	0.012	0.035
2025	0.15	0.000	mid	0.34	0.38	0.16	0.23	0.23	0.23	0.012	0.035
2025	0.15	0.000	high	0.34	0.43	0.21	0.29	0.29	0.29	0.012	0.035
2025	0.2	0.000	low	0.26	0.41	0.18	0.24	0.23	0.23	0.012	0.035
2025	0.2	0.000	mid	0.35	0.37	0.16	0.23	0.23	0.23	0.012	0.035
2025	0.2	0.000	high	0.34	0.43	0.21	0.29	0.29	0.29	0.012	0.035
2026	0.15	0.205	low	0.10	0.65	0.28	0.31	0.28	0.28	0.084	0.211
2026	0.15	0.205	mid	0.09	0.54	0.27	0.30	0.29	0.29	0.084	0.211
2026	0.15	0.205	high	0.10	0.29	0.34	0.34	0.37	0.38	0.084	0.211
2026	0.2	0.088	low	0.09	0.52	0.28	0.30	0.28	0.28	0.080	0.110
2026	0.2	0.088	mid	0.09	0.41	0.28	0.29	0.29	0.29	0.080	0.110
2026	0.2	0.088	high	0.09	0.33	0.35	0.35	0.37	0.38	0.080	0.110

Notes. a^F: the neighborhood benchmark for residents' early conservative share. At s just above s*(1, δ), the allocation attaining the minimum slope. Departures of residents' early composition from a^F are bounded by the station's own departure plus s in every dong. Rows only where the edge exists (s* finite).

Table C6. Least-departure reversal: components and concentration (within-district, calibrated movement)

Election	δ	Residents' early ballots departing	Dongs with > 10	Largest in one dong	Share in top-decile-leverage dongs	Out-of-zone ballots departing	Needed for feasibility	Dongs with > 10	Share in top-decile-leverage dongs
2022	0.1	16,685	31	1,710	95%	5,366	2,015	30	32%
2022	0.2	15,340	31	1,410	94%	6,336	2,015	27	49%
2022	1	8,659	27	892	100%	12,152	2,015	22	71%
2024	0.1	32,177	62	2,218	61%	11,397	8,312	72	5%
2024	0.2	30,103	58	2,218	63%	12,534	8,312	60	11%
2024	1	17,966	55	1,210	62%	23,393	8,312	63	41%
2025	0.1	13,539	26	2,197	96%	3,554	1,063	24	36%
2025	0.2	12,512	26	2,197	96%	4,377	1,063	19	55%
2025	1	8,452	23	1,442	98%	8,162	1,063	21	72%
2026	0.2	69,384	268	1,303	15%	64,253	18,583	244	10%
2026	1	17,423	167	461	20%	59,329	18,583	108	39%

Notes. Departing ballots: absolute departure of residents' conservative ballots from a^F (in-zone early) or from the station anchor (out-of-zone). Needed for feasibility: the smallest out-of-zone departure of any allocation in Lambda, without the slope constraint. Leverage: $|w_i|/H_i$.

Table C7. Restricting how movers vote: residents who voted at another station vote within t of their home station's shares (within-district slope; mobility-calibrated movement, ± 10 points)

t	2022	2024	2025	2026
0.05	**[−0.113, 0.319]**	**[−0.052, 0.260]**	**[−0.106, 0.270]**	[0.054, 0.162]
0.1	**[−0.116, 0.323]**	**[−0.056, 0.263]**	**[−0.111, 0.274]**	[0.051, 0.165]
0.2	**[−0.154, 0.358]**	**[−0.085, 0.292]**	**[−0.151, 0.311]**	[0.042, 0.174]
0.3	**[−0.192, 0.393]**	**[−0.115, 0.321]**	**[−0.186, 0.340]**	[0.033, 0.182]

Table C8. Is the early-vote gap related to price or age? (conservative share; coefficients, standard errors in parentheses)

Election	Gu: in-zone early – election day, on log price	on share aged 60+	Gu: out-of-zone early – election day, on log price	on share aged 60+	Dong (within gu): station departure, on log price	on share aged 60+	Mean gu gap γ^K (SD)
2017	−0.014 (0.008)	0.055 (0.070)	−0.009 (0.011)	−0.037 (0.075)	0.000 (0.005)	0.026 (0.025)	−0.043 (0.008)
2022	−0.001 (0.010)	0.130 (0.076)	−0.042 (0.021)	−0.139 (0.085)	−0.008 (0.006)	0.010 (0.021)	−0.101 (0.010)
2024	−0.021 (0.007)	0.046 (0.059)	−0.029 (0.009)	−0.081 (0.075)	−0.019 (0.006)	−0.010 (0.026)	−0.120 (0.014)
2025	−0.072 (0.012)	−0.179 (0.109)	−0.058 (0.013)	−0.361 (0.110)	−0.011 (0.007)	−0.042 (0.025)	−0.282 (0.024)
2026	−0.025 (0.006)	−0.022 (0.088)	−0.037 (0.010)	−0.257 (0.068)	0.012 (0.007)	−0.010 (0.022)	−0.263 (0.013)

Notes. Gu level: one observation per jurisdiction (48 constituencies in 2024), joint regression on price and age, HC3 standard errors. Dong level: weighted by station ballots, jurisdiction fixed effects, standard errors clustered by jurisdiction. Stations mix residents and movers, so the dong-level test is an imperfect proxy. Age shares from April 2026 resident registration for all elections.

Table C9. The benchmark for out-of-zone voters (within-district slope, mobility-calibrated movement, ±10 points)

Benchmark	Smallest feasible tolerance, median (2022/2024/2025/2026)	2022	2024	2025	2026	2022, + residents' early ballots	2024	2025	2026
station shares (main)	0.025 / 0.020 / 0.062 / 0.044	**[−0.283, 0.448]**	**[−0.185, 0.372]**	**[−0.283, 0.411]**	[0.010, 0.201]	**[−0.026, 0.223]**	[0.001, 0.206]	**[−0.037, 0.194]**	[0.054, 0.156]
shifted by district out-of-zone gap	0.001 / 0.001 / 0.001 / 0.002	**[−0.283, 0.448]**	**[−0.186, 0.373]**	**[−0.283, 0.411]**	[0.007, 0.203]	**[−0.028, 0.223]**	[0.001, 0.208]	**[−0.039, 0.194]**	[0.051, 0.158]
shifted by gap and its price relation	0.001 / 0.001 / 0.001 / 0.002	**[−0.284, 0.448]**	**[−0.186, 0.373]**	**[−0.283, 0.411]**	[0.006, 0.202]	**[−0.035, 0.217]**	[0.001, 0.207]	**[−0.037, 0.196]**	[0.050, 0.157]

Table C10. Robustness of the within-district lower bound (z; $\delta = 0.1$)

Election	Specification	Calibrated, residents' composition unrestricted	Calibrated, s = 0.02	Calibrated, s = 0.05	Unrestricted movement, s = 0.02	Unrestricted movement, s = 0.05
2022	log median (baseline)	**−0.283**	**−0.026**	**−0.051**	**−0.048**	**−0.066**
2022	excl. top 5% by price	**−0.314**	**−0.044**	**−0.070**	**−0.069**	**−0.087**
2022	excl. top 10% by price	**−0.355**	**−0.068**	**−0.095**	**−0.092**	**−0.111**
2022	denominator (H_i) weights	**−0.275**	**−0.011**	**−0.037**	**−0.028**	**−0.048**
2022	log mean	**−0.295**	**−0.024**	**−0.051**	**−0.047**	**−0.067**
2022	linear median	**not signed**	**not signed**	**not signed**	**not signed**	**not signed**
2022	linear mean	**not signed**	**not signed**	**not signed**	**not signed**	**not signed**
2022	excl. dongs with < 30 transactions	**−0.342**	**−0.041**	**−0.069**	**−0.067**	**−0.086**
2024	log median (baseline)	**−0.185**	0.001	**−0.024**	**−0.019**	**−0.037**
2024	excl. top 5% by price	**−0.221**	**−0.018**	**−0.045**	**−0.041**	**−0.060**
2024	excl. top 10% by price	**−0.255**	**−0.041**	**−0.070**	**−0.065**	**−0.085**
2024	denominator (H_i) weights	**−0.153**	0.017	**−0.009**	0.001	**−0.019**
2024	log mean	**−0.189**	0.005	**−0.022**	**−0.015**	**−0.035**
2024	linear median	**not signed**	positive	**not signed**	positive	**not signed**
2024	linear mean	**not signed**	positive	positive	positive	**not signed**
2024	excl. dongs with < 30 transactions	**−0.228**	**−0.010**	**−0.041**	**−0.034**	**−0.057**
2024	constituency fixed effects	**−0.269**	**−0.035**	**−0.066**	**−0.059**	**−0.083**
2025	log median (baseline)	**−0.283**	**−0.037**	**−0.058**	**−0.055**	**−0.069**
2025	excl. top 5% by price	**−0.321**	**−0.056**	**−0.078**	**−0.075**	**−0.090**
2025	excl. top 10% by price	**−0.375**	**−0.087**	**−0.111**	**−0.108**	**−0.124**
2025	denominator (H_i) weights	**−0.277**	**−0.028**	**−0.049**	**−0.041**	**−0.057**
2025	log mean	**−0.286**	**−0.032**	**−0.053**	**−0.049**	**−0.064**
2025	linear median	**not signed**	**not signed**	**not signed**	**not signed**	**not signed**
2025	linear mean	**not signed**	**not signed**	**not signed**	**not signed**	**not signed**

Election	Specification	Calibrated, residents' composition unrestricted	Calibrated, s = 0.02	Calibrated, s = 0.05	Unrestricted movement, s = 0.02	Unrestricted movement, s = 0.05
2025	excl. dongs with < 30 transactions	**−0.342**	**−0.050**	**−0.073**	**−0.071**	**−0.087**
2026	log median (baseline)	0.010	0.054	0.043	0.043	0.033
2026	excl. top 5% by price	**−0.010**	0.041	0.027	0.028	0.016
2026	excl. top 10% by price	**−0.051**	0.008	**−0.007**	**−0.007**	**−0.020**
2026	denominator (H_i) weights	0.030	0.066	0.055	0.057	0.046
2026	log mean	0.016	0.060	0.048	0.048	0.038
2026	linear median	positive	positive	positive	positive	positive
2026	linear mean	positive	positive	positive	positive	positive
2026	excl. dongs with < 30 transactions	0.008	0.055	0.043	0.043	0.031

Notes. Bold marks lower bounds at or below zero. Linear-price slopes are in different units; only their sign is shown.

Table C11. Within-district contrast between high- and low-price dongs (percentage points; z set; y: Proposition 4 lower bound)

Cell	2012	2017	2022	2024	2025	2026
Above vs. at-or-below the gu median price						
conventional	1.9	0.1	4.1	3.8	2.6	3.8
accounting only	**[−1.7, 5.3]**; y ≥ −1.8	**[−16.0, 16.1]**; y ≥ −16.7	**[−43.4, 46.9]**; y ≥ −44.1	**[−30.7, 37.6]**; y ≥ −31.9	**[−26.1, 29.7]**; y ≥ −27.2	**[−19.6, 25.2]**; y ≥ −20.3
$\eta = 0$, $\delta = 1$	**[−1.7, 5.3]**; y ≥ −1.8	**[−5.1, 5.5]**; y ≥ −5.5	**[−8.7, 15.4]**; y ≥ −9.0	**[−5.3, 12.4]**; y ≥ −5.7	**[−7.2, 11.2]**; y ≥ −7.7	**[−6.1, 12.8]**; y ≥ −6.3
$\eta = 0$, $\delta = 0.1$	**[−0.4, 4.1]**; y ≥ 0.0	**[−2.7, 2.9]**; y ≥ −2.8	**[−0.5, 8.5]**; y ≥ −0.3	[0.3, 7.0]; y ≥ 0.5	**[−1.7, 6.6]**; y ≥ −1.8	[1.9, 6.0]; y ≥ 1.7
calibrated, $\delta = 1$	–	–	**[−26.2, 32.6]**; y ≥ −26.5	**[−15.5, 23.0]**; y ≥ −16.1	**[−21.9, 25.6]**; y ≥ −22.7	**[−10.4, 17.0]**; y ≥ −10.8
calibrated, $\delta = 0.1$	–	–	**[−19.2, 26.9]**; y ≥ −19.4	**[−11.7, 19.2]**; y ≥ −12.0	**[−17.8, 22.5]**; y ≥ −18.3	**[−2.4, 10.3]**; y ≥ −2.7
calibrated, $\delta = 0.1$, s = 0.02	–	–	**[−3.2, 11.6]**; y ≥ −3.3	**[−1.6, 9.4]**; y ≥ −1.5	**[−4.2, 9.4]**; y ≥ −4.7	[0.8, 7.2]; y ≥ 0.4
calibrated, $\delta = 0.1$, s = 0.05	–	–	**[−5.1, 13.6]**; y ≥ −5.2	**[−3.0, 10.9]**; y ≥ −3.0	**[−5.9, 10.9]**; y ≥ −6.4	**[0.0, 8.0]**; y ≥ −0.3
Top vs. bottom tercile within gu						
conventional	3.1	0.2	5.4	5.0	3.8	5.4
accounting only	**[−1.0, 6.8]**; y ≥ −1.0	**[−21.7, 21.4]**; y ≥ −22.5	**[−46.6, 52.4]**; y ≥ −47.0	**[−38.2, 45.5]**; y ≥ −39.2	**[−33.5, 39.4]**; y ≥ −34.8	**[−23.9, 31.6]**; y ≥ −24.7
$\eta = 0$, $\delta = 1$	**[−1.0, 6.8]**; y ≥ −1.0	**[−6.9, 7.3]**; y ≥ −7.4	**[−8.9, 18.1]**; y ≥ −9.1	**[−6.6, 15.3]**; y ≥ −7.0	**[−8.4, 15.2]**; y ≥ −9.0	**[−6.5, 16.2]**; y ≥ −6.7
$\eta = 0$, $\delta = 0.1$	[0.3, 5.6]; y ≥ 0.7	**[−3.5, 3.7]**; y ≥ −3.6	**[−0.3, 10.6]**; y ≥ 0.0	[0.7, 8.9]; y ≥ 0.9	**[−1.1, 8.6]**; y ≥ −1.3	[2.9, 8.0]; y ≥ 2.7
calibrated, $\delta = 1$	–	–	**[−26.6, 35.6]**; y ≥ −26.8	**[−18.1, 27.3]**; y ≥ −18.6	**[−26.1, 32.3]**; y ≥ −26.7	**[−11.5, 21.1]**; y ≥ −11.8
calibrated, $\delta = 0.1$	–	–	**[−21.8, 31.4]**; y ≥ −21.9	**[−13.6, 23.0]**; y ≥ −13.8	**[−21.4, 28.0]**; y ≥ −22.0	**[−2.0, 13.1]**; y ≥ −2.4
calibrated, $\delta = 0.1$, s = 0.02	–	–	**[−3.2, 14.0]**; y ≥ −3.2	**[−1.8, 11.9]**; y ≥ −1.6	**[−4.2, 11.6]**; y ≥ −4.7	[1.6, 9.4]; y ≥ 1.2
calibrated, $\delta = 0.1$, s = 0.05	–	–	**[−5.0, 15.9]**; y ≥ −5.1	**[−3.6, 13.7]**; y ≥ −3.5	**[−5.6, 13.0]**; y ≥ −6.3	[0.6, 10.4]; y ≥ 0.2

Table C12. Guaranteed share of the within-district range removed, by reporting change

Election	Assumptions	Current range	All pooled modes by neighborhood	Out-of-zone early only (proposal)	Headcounts only	Both
2017	minimal movement	0.298	97%	85%	0%	22%
2022	minimal movement	0.364	92%	76%	0%	10%
2022	half the mobility-calibrated movement	0.692	29%	24%	0%	10%
2022	mobility-calibrated movement	0.841	20%	16%	0%	10%
2022	mobility-calibrated movement, residents resemble	0.591	82%	69%	6%	89%
2022	mobility-calibrated movement, ±10 points	0.731	8%	4%	3%	7%
2022	mobility-calibrated movement, ±10 points, residents resemble	0.249	57%	27%	3%	70%
2024	minimal movement	0.331	92%	83%	0%	12%
2024	half the mobility-calibrated movement	0.577	34%	30%	0%	12%
2024	mobility-calibrated movement	0.676	25%	22%	0%	12%
2024	mobility-calibrated movement, residents resemble	0.508	79%	72%	13%	88%
2024	mobility-calibrated movement, ±10 points	0.557	9%	6%	4%	8%
2024	mobility-calibrated movement, ±10 points, residents resemble	0.205	49%	30%	6%	63%
2025	minimal movement	0.335	94%	80%	0%	21%
2025	half the mobility-calibrated movement	0.665	28%	23%	0%	21%
2025	mobility-calibrated movement	0.787	20%	18%	0%	21%
2025	mobility-calibrated movement, residents resemble	0.532	83%	70%	16%	88%
2025	mobility-calibrated movement, ±10 points	0.694	9%	6%	6%	15%
2025	mobility-calibrated movement, ±10 points, residents resemble	0.231	61%	32%	4%	68%
2026	minimal movement	0.280	99%	95%	1%	44%
2026	half the mobility-calibrated movement	0.379	74%	71%	1%	44%
2026	mobility-calibrated movement	0.404	69%	66%	1%	44%
2026	mobility-calibrated movement, residents resemble	0.347	91%	87%	4%	93%

Election	Assumptions	Current range	All pooled modes by neighborhood	Out-of-zone early only (proposal)	Headcounts only	Both
2026	mobility-calibrated movement, ±10 points	0.191	33%	27%	10%	23%
2026	mobility-calibrated movement, ±10 points, residents resemble	0.101	69%	57%	7%	72%

Notes. Out-of-zone voters are unrestricted unless ±10 points is stated. All pooled modes and Out-of-zone early only: share of the current range in the same row removed. Headcounts only and Both: share removed of the range under the same composition assumptions but with movement unrestricted, since a headcount report replaces the movement assumption. Whether a report reveals the direction of the slope: Table C16.

Table C13. Stability of headline sets to resampling prices and dongs (z, $\delta = 0.1$)

Election	Slope	Movement	Slack s	Identified set	SD of bounds, transactions resampled	SD of bounds, dongs and transactions resampled	[5th pct of lower, 95th pct of upper]	Replications with positive lower bound	Replications
2022	Within-district	calibrated	0.02	**[−0.026, 0.223]**	0.002, 0.004	0.010, 0.012	[−0.078, 0.268]	0%	200
2022	Citywide	calibrated	0.02	[0.080, 0.216]	0.002, 0.003	0.008, 0.010	[0.034, 0.256]	100%	200
2022	Citywide	calibrated	–	**[−0.062, 0.340]**	0.002, 0.004	0.013, 0.015	[−0.163, 0.422]	0%	200
2024	Within-district	calibrated	–	**[−0.185, 0.372]**	0.004, 0.008	0.017, 0.020	[−0.280, 0.453]	0%	200
2024	Within-district	calibrated	0.02	[0.001, 0.206]	0.002, 0.004	0.010, 0.011	[−0.042, 0.242]	0%	200
2024	Within-constituency	calibrated	0.02	**[−0.035, 0.221]**	0.002, 0.006	0.011, 0.013	[−0.082, 0.257]	0%	200
2024	Citywide	calibrated	0.02	[0.102, 0.194]	0.002, 0.002	0.007, 0.007	[0.063, 0.224]	100%	200
2024	Citywide	calibrated	–	[0.018, 0.268]	0.002, 0.003	0.010, 0.010	[−0.068, 0.340]	0%	200
2026	Within-district	calibrated	–	[0.010, 0.201]	0.002, 0.002	0.009, 0.009	[−0.031, 0.237]	4%	200
2026	Within-district	calibrated	0.02	[0.054, 0.156]	0.001, 0.002	0.007, 0.007	[0.030, 0.176]	100%	200
2026	Within-district	calibrated	0.05	[0.043, 0.168]	0.001, 0.002	0.008, 0.008	[0.015, 0.191]	100%	200
2026	Within-district	unrestricted	0.02	[0.043, 0.168]	0.001, 0.002	0.008, 0.008	[0.011, 0.194]	100%	200
2026	Citywide	calibrated	0.02	[0.111, 0.154]	0.001, 0.001	0.006, 0.005	[0.088, 0.172]	100%	200
2026	Citywide	calibrated	–	[0.092, 0.174]	0.001, 0.001	0.007, 0.006	[0.054, 0.205]	100%	200

Notes. The returns, rolls and movement caps, which define the polytope, are held fixed. Transactions resampled: each dong's median price is recomputed from a resample of its transactions; this measures sensitivity to price measurement. Dongs and transactions resampled: dongs are also resampled with replacement within jurisdictions, which duplicates high-leverage dongs and mechanically pushes the extrema outward, so these columns overstate instability and are not confidence intervals.

Table C14. Regressions across fully counted units (valid-vote share on the unit mean of dong log price)

Election	Units	Slope	HC1 SE	HC3 SE	Permutation p	Leave-one-out range	Excl. Gangnam, Seocho and Songpa: slope (HC3 SE)	Electorate-weighted price: slope (HC3 SE)
2012	25 gu	0.137	0.028	0.035	<0.001	0.111–0.148	0.057 (0.038)	0.138 (0.035)
2017	25 gu	0.060	0.014	0.017	<0.001	0.048–0.065	0.025 (0.025)	0.061 (0.018)
2022	25 gu	0.197	0.028	0.033	<0.001	0.175–0.207	0.131 (0.022)	0.203 (0.034)
2024	24 units	0.172	0.033	0.038	<0.001	0.151–0.193	0.097 (0.039)	0.175 (0.038)
2024	48 constituencies	0.181	0.023	0.025	<0.001	0.168–0.194	0.099 (0.025)	0.183 (0.025)
2025	25 gu	0.137	0.023	0.028	<0.001	0.118–0.147	0.080 (0.022)	0.139 (0.028)
2026	25 gu	0.142	0.026	0.030	<0.001	0.123–0.153	0.086 (0.028)	0.145 (0.031)

Notes. 2024: Jung-gu and Seongdong-gu merged because one constituency spans both. Slopes describe the fully counted units themselves and need no allocation assumption. The standard errors and permutation p-values (5,000 permutations) answer a different question: how far the slope would vary if district values were draws from a larger population or if prices were assigned to districts at random. They are reported for that reading only; the identified sets of the main analysis condition on the observed returns and involve no sampling.

Table C15. Valid-vote shares: within-district slope

Election	Assumptions	Roll-based share (exact range)	Valid-vote share, lower end: [outer bound, attainable]	upper end: [attainable, outer bound]
2012	returns and rolls only	[0.008, 0.157]	[0.0069, 0.0070]	[0.1577, 0.1579]
2012	minimal movement	[0.008, 0.157]	[0.0069, 0.0070]	[0.1577, 0.1579]
2012	minimal movement, ±10 points	[0.027, 0.140]	[0.0356, 0.0359]	[0.1323, 0.1326]
2017	returns and rolls only	**[−0.428, 0.450]**	[−0.4405, −0.4368]	[0.4577, 0.4618]
2017	minimal movement	**[−0.136, 0.162]**	[−0.1440, −0.1422]	[0.1676, 0.1699]
2017	minimal movement, ±10 points	**[−0.060, 0.085]**	[−0.0603, −0.0597]	[0.0857, 0.0863]
2022	returns and rolls only	**[−0.615, 0.704]**	[−0.6208, −0.6195]	[0.7098, 0.7115]
2022	minimal movement	**[−0.100, 0.264]**	[−0.1016, −0.1013]	[0.2658, 0.2663]
2022	minimal movement, ±10 points	[0.014, 0.175]	[0.0187, 0.0203]	[0.1686, 0.1697]
2022	mobility-calibrated movement, ±10 points	**[−0.283, 0.448]**	[−0.2849, −0.2832]	[0.4484, 0.4502]
2022	+ residents' early ballots resemble neighborhood	**[−0.026, 0.223]**	[−0.0258, −0.0237]	[0.2187, 0.2200]
2024	returns and rolls only	**[−0.553, 0.683]**	[−0.5671, −0.5621]	[0.6937, 0.6971]
2024	minimal movement	**[−0.077, 0.254]**	[−0.0829, −0.0817]	[0.2582, 0.2592]
2024	minimal movement, ±10 points	[0.033, 0.164]	[0.0362, 0.0374]	[0.1587, 0.1594]
2024	mobility-calibrated movement, ±10 points	**[−0.185, 0.372]**	[−0.1883, −0.1851]	[0.3734, 0.3759]
2024	+ residents' early ballots resemble neighborhood	[0.001, 0.206]	[0.0045, 0.0055]	[0.2027, 0.2039]
2025	returns and rolls only	**[−0.468, 0.552]**	[−0.4821, −0.4784]	[0.5612, 0.5639]
2025	minimal movement	**[−0.104, 0.232]**	[−0.1099, −0.1089]	[0.2358, 0.2367]
2025	minimal movement, ±10 points	[0.002, 0.149]	[−0.0021, −0.0007]	[0.1512, 0.1523]
2025	mobility-calibrated movement, ±10 points	**[−0.283, 0.411]**	[−0.2898, −0.2876]	[0.4160, 0.4178]
2025	+ residents' early ballots resemble neighborhood	**[−0.037, 0.194]**	[−0.0459, −0.0445]	[0.2010, 0.2024]
2026	returns and rolls only	**[−0.279, 0.439]**	[−0.2874, −0.2854]	[0.4425, 0.4442]

Election	Assumptions	Roll-based share (exact range)	Valid-vote share, lower end: [outer bound, attainable]	upper end: [attainable, outer bound]
2026	minimal movement	**[−0.046, 0.234]**	[−0.0489, −0.0487]	[0.2351, 0.2354]
2026	minimal movement, ±10 points	[0.072, 0.138]	[0.0696, 0.0698]	[0.1389, 0.1390]
2026	mobility-calibrated movement, ±10 points	[0.010, 0.201]	[0.0063, 0.0070]	[0.2027, 0.2033]
2026	+ residents' early ballots resemble neighborhood	[0.054, 0.156]	[0.0493, 0.0496]	[0.1588, 0.1591]

Notes. The exact lower end of the valid-vote range lies between the two numbers given; it is negative, so the direction is not established, when the attainable value is negative.

Table C16. Opposite-signed slopes compatible with the same report: maximum margin t*

Election	Assumptions	Residence reporting	Out-of-zone only	Residence + headcounts (no movement assumption)
2022	mobility-calibrated movement	0.337	0.342	0.594
2022	+ residents' early ballots (s = 0.02)	0.053	0.090	0.042
2022	+ out-of-zone ±10 points	0.283	0.283	0.591
2022	main (Table 2, column E)	0.026	0.026	0.032
2024	mobility-calibrated movement	0.248	0.248	0.545
2024	+ residents' early ballots (s = 0.02)	0.052	0.071	0.043
2024	+ out-of-zone ±10 points	0.185	0.185	0.527
2024	main (Table 2, column E)	revealed	revealed	0.019
2025	mobility-calibrated movement	0.315	0.325	0.404
2025	+ residents' early ballots (s = 0.02)	0.045	0.079	0.041
2025	+ out-of-zone ±10 points	0.283	0.283	0.404
2025	main (Table 2, column E)	0.035	0.037	0.033
2026	mobility-calibrated movement	0.064	0.069	0.200
2026	+ residents' early ballots (s = 0.02)	0.016	0.022	0.018
2026	+ out-of-zone ±10 points	revealed	revealed	0.168
2026	main (Table 2, column E)	revealed	revealed	revealed

Notes. t* = max t subject to $\theta(\ell) \le -t$, $\theta(\ell') \ge t$, $S\ell = S\ell'$, with ℓ and ℓ' in Λ. A report reveals the direction iff no two allocations it cannot distinguish have slopes of strictly opposite sign, that is iff t* ≤ 0 ("revealed"). Where the current set already excludes zero every report reveals the direction; the last column drops the movement assumption, as in Online Appendix Table C12.

Table C17. Beyond Gangnam: the within-district relation without Gangnam, Seocho and Songpa

Election	Conv.	(A) Returns and rolls only	(B) Minimal movement	(C) = (B) + out-of-zone within 10 points	(D) Mobility-calibrated	(E) Main
A. Within-district slope without Gangnam, Seocho and Songpa (z)						
2012	0.058	**[−0.021, 0.129]**	**[−0.021, 0.129]**	[0.001, 0.110]	–	–
2017	0.000	**[−0.471, 0.441]**	**[−0.158, 0.150]**	**[−0.074, 0.071]**	–	–
2022	0.076	**[−0.656, 0.716]**	**[−0.129, 0.253]**	**[−0.007, 0.157]**	**[−0.300, 0.432]**	**[−0.042, 0.204]**
2024	0.075	**[−0.602, 0.689]**	**[−0.112, 0.235]**	[0.001, 0.140]	**[−0.224, 0.357]**	**[−0.027, 0.184]**
2025	0.051	**[−0.523, 0.557]**	**[−0.140, 0.216]**	**[−0.026, 0.127]**	**[−0.317, 0.400]**	**[−0.064, 0.173]**
2026	0.082	**[−0.315, 0.446]**	**[−0.079, 0.228]**	[0.047, 0.119]	**[−0.020, 0.188]**	[0.029, 0.139]
B. Higher- minus lower-price half without the three (z, points)						
2012	0.9	**[−2.6, 4.1]**	**[−2.6, 4.1]**	**[−1.3, 2.9]**	–	–
2017	−0.4	**[−16.1, 15.1]**	**[−5.4, 4.8]**	**[−3.1, 2.3]**	–	–
2022	3.1	**[−43.9, 46.6]**	**[−9.6, 14.7]**	**[−1.4, 7.6]**	**[−19.5, 25.6]**	**[−3.9, 10.5]**
2024	2.7	**[−32.0, 37.0]**	**[−6.5, 11.2]**	**[−0.8, 5.9]**	**[−13.0, 18.4]**	**[−2.6, 8.3]**
2025	1.7	**[−26.5, 28.9]**	**[−7.9, 10.3]**	**[−2.6, 5.6]**	**[−18.4, 21.4]**	**[−5.1, 8.4]**
2026	2.8	**[−20.2, 24.1]**	**[−7.0, 11.7]**	[0.8, 5.0]	**[−3.6, 9.3]**	**[−0.3, 6.2]**

Notes. Columns as in Table 2; roll-based share z. Valid-vote shares: Table C18.

Table C18. Valid-vote slope lower bounds with and without the three southeastern districts

Election	Slope: all	Slope: without the three	Contrast (points): all	Contrast (points): without the three
2022	[−0.026, −0.024]	[−0.044, −0.042]	[−3.3, −3.2]	[−4.0, −3.9]
2024	[0.004, 0.005]	[−0.026, −0.025]	[−1.5, −1.4]	[−2.7, −2.6]
2025	[−0.046, −0.045]	[−0.072, −0.071]	[−4.7, −4.6]	[−5.6, −5.5]
2026	[0.049, 0.050]	[0.024, 0.024]	[0.4, 0.5]	[−0.6, −0.6]

Notes. Each entry gives a guaranteed outer bound and an attainable value for the lower end of the valid-vote range; the exact lower end lies between them. The direction is established when the first value is positive and not established when the second is negative.

Table C19. Sensitivity thresholds s* with and without the three southeastern districts

Election	Movement	Out-of-zone tolerance (points)	Slope: all	Slope: without the three	Contrast: all	Contrast: without the three
2022	mobility-calibrated	±10	none	none	none	none
2022	mobility-calibrated	±20	none	none	none	none
2022	unrestricted	±10	none	none	none	none
2022	unrestricted	±20	none	none	none	none
2024	mobility-calibrated	±10	0.021	none	none	none
2024	mobility-calibrated	±20	none	none	none	none
2024	unrestricted	±10	none	none	none	none
2024	unrestricted	±20	none	none	none	none
2025	mobility-calibrated	±10	none	none	none	none
2025	mobility-calibrated	±20	none	none	none	none
2025	unrestricted	±10	none	none	none	none
2025	unrestricted	±20	none	none	none	none
2026	mobility-calibrated	±10	∞	0.103	0.049	0.010
2026	mobility-calibrated	±20	0.087	0.001	none	none
2026	unrestricted	±10	0.122	0.067	0.019	none
2026	unrestricted	±20	0.006	none	none	none

Notes. s*: the largest slack s at which the lower end of the range of the within-district slope (or of the higher-minus-lower-price contrast) is positive, with movement and the out-of-zone tolerance held fixed; bisection width 0.0012, lower end reported. The main setting uses s = 0.02, so a relation is established there when s* exceeds 0.02. ∞: positive however much residents' early ballots depart; none: not positive for any s.

Table C20. Point-identified benchmarks: election-day voters and in-zone early stations

Election	Election-day voters: all	Election-day voters: without the three	In-zone stations: all	In-zone stations: without the three
A. Within-district slope				
2012	0.087	0.057	–	–
2017	0.013	0.001	0.014	−0.004
2022	0.097	0.079	0.091	0.066
2024	0.102	0.078	0.090	0.058
2025	0.063	0.041	0.055	0.029
2026	0.086	0.069	0.102	0.073
B. Higher- minus lower-price half (percentage points)				
2012	1.9	0.9	–	–
2017	0.1	−0.4	0.2	−0.4
2022	4.0	3.2	3.8	2.8
2024	3.8	2.7	3.3	2.1
2025	1.8	1.0	1.8	0.9
2026	3.0	2.2	3.8	2.5
C. Auxiliary diagnostics for the election-day slope: gu-clustered standard error; within-district permutation p-value				
2012	(0.021); p <0.001	(0.020); p <0.001		
2017	(0.011); p 0.096	(0.012); p 0.875		
2022	(0.017); p <0.001	(0.016); p <0.001		
2024	(0.016); p <0.001	(0.013); p <0.001		
2025	(0.015); p <0.001	(0.014); p <0.001		
2026	(0.012); p <0.001	(0.012); p <0.001		

Notes. Conservative candidate's valid-vote share. Election-day ballots are counted in the voter's own voting district, so these slopes and contrasts describe registered residents who voted on election day and need no allocation assumption. In-zone station shares are counted where cast and include voters from other neighborhoods of the zone; 2012 had no in-zone early voting. Weights as in Table 2 (gu fixed effects; median split within gu). Without the three: Gangnam, Seocho and Songpa excluded from the weights. Panel C, auxiliary diagnostics only: the slopes above describe the observed neighborhoods exactly. The permutation p-value (prices permuted within each gu, 5,000 permutations, two-sided) assumes that dongs are exchangeable within a gu and ignores spatial dependence; prices were not randomly assigned. The clustered standard error reads the dongs as draws from a larger population.

Table C21. Out-of-zone and in-zone early voters by counting jurisdiction (conservative valid-vote share, percentage points)

Election	Jurisdictions	Out-of-zone minus in-zone: median	Range	Largest absolute gap	In-zone stations: median absolute deviation from their jurisdiction	Largest station deviation
2017	25	−3.5	−5.4 to −0.6	5.4	1.7	14.8
2022	25	1.7	−3.2 to 3.4	3.4	2.2	19.9
2024	48	−2.5	−6.0 to 0.1	6.0	1.8	19.2
2025	25	0.4	−2.3 to 2.7	2.7	1.4	21.9
2026	25	3.6	1.7 to 7.7	7.7	1.9	25.7

Notes. Each jurisdiction's out-of-zone early ballots are published as one total, so the first three columns describe whole districts (constituencies in 2024). The tolerance δ bounds the same difference for each neighborhood's residents, which is not observed and may vary more within a district than these district-level gaps; the station columns show how far neighborhoods' in-zone shares depart from their district's in-zone share. These are references for δ, not tests of it.

Table C22. How far residents' early-vote departures may align with price before the within-district slope loses its sign

Election	κ^*: all, out-of-zone ± 10	κ^*: all, out-of-zone unrestricted	κ^*: without the three, ± 10	κ^*: without the three, unrestricted	Station gradient: median	Station gradient: largest absolute	Between-jurisdiction reference
2022	0.042	0.042	0.028	0.028	−0.003	0.016	0.007
2024	0.052	0.038	0.015	<0.001	−0.003	0.047	0.008
2025	0.032	0.032	0.011	0.011	−0.001	0.026	0.021
2026	∞	0.095	0.091	0.075	0.004	0.013	0.006

Notes. Mobility-calibrated movement; the restriction on residents' early-vote composition (s) is replaced by $|g_u| \leq \kappa$, where g_u is the slope, within counting jurisdiction u, of residents' early-vote departures from their benchmarks (in-zone and out-of-zone, as a share of the roll) on log price. κ^*: the largest κ at which the lower end of the within-district slope range stays positive (bisection width 0.0002; ∞: positive without the restriction). Station gradient: the same slope computed for observed in-zone station ballots, over jurisdictions; stations also count non-residents. Between-jurisdiction reference: the slope residents' departures would have if, within jurisdictions, early-vote gaps varied with price as they do across counting jurisdictions (gu; National Assembly constituencies in 2024), $|b_K|$ (in-zone early ballots / H) + $|b_B - b_K|$ (out-of-zone ballots / H), with b_K and b_B the cross-district slopes of the in-zone and out-of-zone early minus election-day gaps on log price. The slope is identified as positive when residents' departures are no more aligned with price than κ^*.

Table C23. Within-district relation with age composition held fixed, and the tercile contrast

Election	All of Seoul: election-day voters	All of Seoul: all residents, main assumptions	Without the three: election-day voters	Without the three: all residents, main assumptions
Slope				
2022	0.097	**[−0.026, 0.223]**	0.079	**[−0.042, 0.204]**
2024	0.102	[0.001, 0.206]	0.078	**[−0.027, 0.184]**
2025	0.063	**[−0.037, 0.194]**	0.041	**[−0.064, 0.173]**
2026	0.086	[0.054, 0.156]	0.069	[0.029, 0.139]
Slope, age held fixed				
2022	0.087	**[−0.052, 0.227]**	0.071	**[−0.064, 0.210]**
2024	0.100	**[−0.017, 0.214]**	0.081	**[−0.040, 0.199]**
2025	0.067	**[−0.051, 0.208]**	0.049	**[−0.071, 0.192]**
2026	0.088	[0.046, 0.160]	0.074	[0.024, 0.148]
Top minus bottom tercile (points)				
2022	5.4	**[−3.2, 14.0]**	4.0	**[−4.2, 12.5]**
2024	4.9	**[−1.8, 11.9]**	3.5	**[−3.1, 10.4]**
2025	2.9	**[−4.2, 11.6]**	1.6	**[−5.4, 10.0]**
2026	4.4	[1.6, 9.4]	3.1	[0.1, 7.9]

Notes. Election-day voters: valid-vote share, point-identified. All residents: roll-based share z, mobility-calibrated movement, out-of-zone within 10 points, s = 0.02. Age held fixed: slope weights from a regression of log price on gu dummies, the mean age of residents aged 18 and over and the share aged 60 and over (Ministry of the Interior and Safety, April 2026; a proxy for earlier elections). Tercile: mean of each gu's top third by price minus its bottom third, gu weights as for the median split.

Table C24. Alternative outcomes (z)

Election	Outcome	Slope	Conventional	Returns and rolls only	Minimal movement (η = 0)
2012	Democratic	Citywide	−0.122	[−0.152, −0.088]	[−0.152, −0.088]
2012	Democratic	Within-district	−0.086	[−0.157, −0.007]	[−0.157, −0.007]

Election	Outcome	Slope	Conventional	Returns and rolls only	Minimal movement ($\eta = 0$)
2017	Hong+Yoo	Citywide	0.085	**[−0.139, 0.299]**	[0.002, 0.160]
2017	Hong+Yoo+Ahn	Citywide	0.066	**[−0.206, 0.323]**	**[−0.037, 0.162]**
2017	Democratic	Citywide	−0.056	**[−0.314, 0.217]**	**[−0.154, 0.047]**
2017	Hong+Yoo	Within-district	0.041	**[−0.500, 0.556]**	**[−0.154, 0.225]**
2017	Hong+Yoo+Ahn	Within-district	0.030	**[−0.624, 0.657]**	**[−0.212, 0.264]**
2017	Democratic	Within-district	−0.023	**[−0.654, 0.633]**	**[−0.260, 0.221]**
2022	Democratic	Citywide	−0.138	**[−0.482, 0.251]**	[−0.232, −0.031]
2022	Democratic	Within-district	−0.087	**[−0.713, 0.618]**	**[−0.260, 0.105]**
2024	PPP+Reform	Citywide	0.153	**[−0.147, 0.408]**	[0.070, 0.218]
2024	Democratic	Citywide	−0.144	**[−0.408, 0.156]**	[−0.214, −0.061]
2024	PPP+Reform	Within-district	0.106	**[−0.553, 0.686]**	**[−0.075, 0.257]**
2024	Democratic	Within-district	−0.098	**[−0.694, 0.565]**	**[−0.258, 0.086]**
2025	Kim+Lee	Citywide	0.128	**[−0.141, 0.363]**	[0.036, 0.206]
2025	Democratic	Citywide	−0.126	**[−0.368, 0.148]**	[−0.206, −0.033]
2025	Kim+Lee	Within-district	0.087	**[−0.538, 0.633]**	**[−0.121, 0.270]**
2025	Democratic	Within-district	−0.085	**[−0.645, 0.552]**	**[−0.273, 0.126]**
2026	Democratic	Citywide	−0.123	**[−0.273, 0.047]**	[−0.180, −0.058]
2026	Democratic	Within-district	−0.094	**[−0.446, 0.301]**	**[−0.231, 0.057]**

Notes. 2017 Hong+Yoo: the two conservative candidates combined (Hong+Yoo+Ahn adds the centrist Ahn Cheol-soo). 2024 PPP+Reform and 2025 Kim+Lee: the main conservative candidate with the minor conservative candidate.

Table C25. Further restrictions: the out-of-zone tolerance, movers' composition and price-aligned out-of-zone departures (within-district slope, z)

Election	δ* at s = 0.02: all	δ*: without the three	s = 0.02 and movers within 10 points: all	Same, without the three	κ_P*: all	κ_P*: without the three	U, share of H
2022	0.037	none	**[−0.025, 0.222]**	**[−0.042, 0.203]**	0.007	none	0.33%
2024	0.102	none	[0.003, 0.202]	**[−0.026, 0.179]**	0.015	none	0.40%
2025	none	none	**[−0.037, 0.193]**	**[−0.063, 0.172]**	<0.001	none	0.25%
2026	0.261	0.176	[0.060, 0.153]	[0.033, 0.135]	0.082	0.060	0.03%

Notes. Mobility-calibrated movement throughout. δ*: the largest out-of-zone tolerance at which the lower end of the range stays positive with s = 0.02 (none: not positive at the smallest feasible tolerance). Movers: residents who voted at another station of their zone vote within 10 points of their home station's composition, with s = 0.02 and δ = 0.1 (5 points gives nearly the same ranges). κ_P*: with δ removed and s = 0.02, the largest bound on the within-jurisdiction slope of residents' out-of-zone departures from their station's composition on log price at which the slope stays positive. U: removed residents who cast no counted ballot (allocated without restriction), as a share of H.

D. A published allocation

A polling firm's research note on the 2026 local elections (Lee 2026) reported dong-level correlations between apartment prices and Oh Se-hoon's share in Seoul of 0.84 citywide and 0.71 within gu, and a within-gu slope of about 1.9 points per 10 million won per pyeong. We followed its description: residents' pooled-mode counts from the roll difference, with at-home counts in proportion to registered at-home voters; out-of-zone ballots at the dong's in-zone early composition; at-home ballots at the dong's overall composition; columns rescaled to the gu × candidate totals; integers by largest remainder; mean price per pyeong over June 2025 to May 2026. On 423 dongs (424 in the original) this gives 0.842, 0.709 and 1.88, matching the reported figures to their precision.

The replication permits three separate checks. The allocation reproduces the official gu totals by construction. Its premise that residents' out-of-zone ballots have their station's composition cannot hold exactly, because out-of-zone early voters were more favorable to Oh than in-zone early voters in every gu (median 3.6 points, Table C21); the rescaling changes the premise. The allocation does not satisfy every accounting restriction: its constituency-level out-of-zone headcounts are within a median 0.4% of the council returns' counts, but in 47 of 155 constituencies they exceed what the roll leaves for pooled ballots and non-voting, by a median of 16 voters (Table D1). Its slope of 1.88 lies inside every identified set in Table D2, including the set under the paper's main assumptions ([1.16, 2.54]), so its positive slope is compatible with the range identified under the main assumptions in 2026.

Table D1. Admissibility of the published 2026 allocation

Gu	Smallest η making it feasible, movement anywhere in the gu	Constituencies where it assigns more pooled voters than the roll allows	Largest excess (voters)	Smallest station-anchor band, η = 10	Out-of-zone minus in-zone early conservative share
Gangnam-gu	0.023	5 of 9	56	0.004	+0.044
Gangdong-gu	0.005	1 of 7	73	0.008	+0.035
Gangbuk-gu	0.000	0 of 4	–	0.023	+0.023
Gangseo-gu	0.010	3 of 8	30	0.042	+0.044
Gwanak-gu	0.008	3 of 7	28	0.056	+0.050

Gu	Smallest η making it feasible, movement anywhere in the gu	Constituencies where it assigns more pooled voters than the roll allows	Largest excess (voters)	Smallest station-anchor band, η = 10	Out-of-zone minus in-zone early conservative share
Gwangjin-gu	0.005	1 of 4	21	0.021	+0.040
Guro-gu	0.006	2 of 6	2	0.032	+0.035
Geumcheon-gu	0.001	1 of 4	5	0.031	+0.029
Nowon-gu	0.004	2 of 6	3	0.047	+0.055
Dobong-gu	0.002	1 of 5	4	0.021	+0.016
Dongdaemun-gu	0.010	2 of 6	79	0.035	+0.040
Dongjak-gu	0.007	1 of 7	72	0.021	+0.026
Mapo-gu	0.010	4 of 8	37	0.039	+0.042
Seodaemun-gu	0.005	1 of 5	477	0.029	+0.031
Seocho-gu	0.005	3 of 5	38	0.004	+0.023
Seongdong-gu	0.009	1 of 4	16	0.031	+0.053
Seongbuk-gu	0.005	1 of 7	21	0.029	+0.036
Songpa-gu	0.022	4 of 10	43	0.005	+0.032
Yangcheon-gu	0.005	3 of 8	87	0.021	+0.036
Yeongdeungpo-gu	0.011	2 of 7	27	0.015	+0.037
Yongsan-gu	0.013	2 of 5	51	0.026	+0.027
Eunpyeong-gu	0.003	1 of 8	2	0.031	+0.029
Jongno-gu	0.004	1 of 4	126	0.052	+0.076
Jung-gu	0.016	1 of 4	5	0.010	+0.027
Jungnang-gu	0.006	1 of 7	278	0.029	+0.024

Notes. Feasibility fixes the allocation's dong-level pooled ballots and asks whether some in-zone movement and non-voting reproduce (R) and (S). Under the statutory zones, a constituency's pooled voters plus non-voters must equal its removed residents minus its in-zone early ballots.

Table D2. The published 2026 allocation and the identified range (within-district slope, percentage points per 10 million won per pyeong)

Specification	Movement	Out-of-zone tolerance	Range consistent with data	Published allocation
gu FE	returns and rolls only	none	**[−3.29, 6.45]**	1.88
gu FE + age	returns and rolls only	none	**[−3.30, 6.45]**	1.88
gu FE	$\eta = 0$	none	**[−0.12, 3.64]**	1.88
gu FE + age	$\eta = 0$	none	**[−0.13, 3.64]**	1.88
gu FE	$\eta = 0$	±10	[1.44, 2.31]	1.88
gu FE + age	$\eta = 0$	±10	[1.44, 2.31]	1.88
gu FE	$\eta = 0$	±20	[1.05, 2.70]	1.88
gu FE + age	$\eta = 0$	±20	[1.04, 2.70]	1.88
gu FE	mobility-calibrated movement	±10	[0.60, 3.17]	1.88
gu FE + age	mobility-calibrated movement	±10	[0.60, 3.17]	1.88
gu FE	mobility-calibrated movement	none	**[−0.98, 4.49]**	1.88
gu FE + age	mobility-calibrated movement	none	**[−0.98, 4.49]**	1.88
gu FE	mobility-calibrated movement, residents resemble	±10	[1.16, 2.54]	1.88
gu FE + age	mobility-calibrated movement, residents resemble	±10	[1.16, 2.55]	1.88
gu FE	mobility-calibrated movement, residents resemble	none	**[−0.63, 4.08]**	1.88
gu FE + age	mobility-calibrated movement, residents resemble	none	**[−0.64, 4.09]**	1.88

References

Amos, Brian, McDonald, Michael P. and Watkins, Russell (2017) When boundaries collide: Constructing a national database of demographic and voting statistics. *Public Opinion Quarterly* 81(S1), 385-400. https://doi.org/10.1093/poq/nfx001

Australian Electoral Commission (2025) *2025 Federal Election Report*. Canberra: AEC. Available at: https://www.aec.gov.au/Elections/federal_elections/2025/2025-Federal-Election-Report.pdf (accessed 11 October 2026).

Canada (2000) *Canada Elections Act*, S.C. 2000, c. 9, s. 168. Available at: https://laws-lois.justice.gc.ca/eng/acts/E-2.01/ (accessed 11 October 2026).

Die Bundeswahlleiterin (2025) *Wahl zum 21. Deutschen Bundestag am 23. Februar 2025*, Heft 5, Teil 2. Wiesbaden. Available at: https://bundeswahlleiterin.de/ (accessed 11 October 2026).

Dinkelbach, Werner (1967) On nonlinear fractional programming. *Management Science* 13(7), 492-498. https://doi.org/10.1287/mnsc.13.7.492

Elections Canada (2025) *Forty-Fifth General Election 2025: Official Voting Results*, Table 5. Available at: https://www.elections.ca/res/rep/off/ovrGE45/62/table5E.html (accessed 11 October 2026).

Germany (2002) *Bundeswahlordnung* [Federal Electoral Regulations], as promulgated 19 April 2002, § 7. Available at: https://www.gesetze-im-internet.de/bwo_1985/ (accessed 11 October 2026).

Japan (1950) 公職選挙法 [Public Offices Election Act], Act No. 100 of 1950, Articles 18 and 66(2). e-Gov Laws. Available at: https://laws.e-gov.go.jp/law/325AC1000000100 (accessed 10 October 2026).

Lee, Dong-han (2026) [9hoe jiseon teukjip] budongsangwa sudogwon eummyeondong pyosim bunseok (Ninth local elections: Real estate and voting patterns in the capital region's eups, myeons and dongs). *Hankook Research: Public Opinion within Public Opinion*, 10 June. Research note [in Korean]. Available at: https://hrcopinion.co.kr/archives/36779 (accessed 10 October 2026).

Ministry of Land, Infrastructure and Transport (2026) Apartment sale transactions, detailed records (RTMSDataSvcAptTradeDev), monthly records for Seoul in the twelve months before each election studied (December 2011 to May 2026). Public Data Portal [in Korean]. Available at: https://www.data.go.kr/ (accessed 10 October 2026).

National Election Commission (2026) *Election Statistics System* and *Public Data Portal* files: counting-unit returns (중앙선거관리위원회_대통령선거개표결과, files for 19 December 2012, 9 May 2017, 9 March 2022 and 3 June 2025; 중앙선거관리위원회_국회의원선거개표결과, 10 April 2024); counting-unit spreadsheets for the 3 June 2026 Seoul mayoral election, by district; certified electorates by dong (ElcntInfoInqireService, getEmdElcntInfoInqire, data.go.kr); the table of constituencies and dongs (선거구및읍면동현황) and district-council returns for 3 June 2026. Available at: https://info.nec.go.kr/ and https://www.data.go.kr/ (accessed 11 October 2026).

National Election Commission (n.d.) Regulations on Public Official Elections, Articles 86(3) and 133(3), in the versions in force at each election studied; Article 86(3) as amended 24 December 2015 and Article 133(3) as amended 9 March 2018. Korean Law Information Center [in Korean]. Available at: https://www.law.go.kr/%EB%B2%95%EB%A0%B9/%EA%B3%B5%EC%A7%81%EC%84%A0%EA%B1%B0%EA%B4%80%EB%A6%AC%EA%B7%9C%EC%B9%99 (accessed 12 October 2026).

Palmer, Maxwell and Schneer, Benjamin (2021) *Racially Polarized Voting Analysis for the Virginia Redistricting Commission*. Report, 31 August. Available at: https://virginiaredistricting.org/2021/Data/Publications/palmer_schneer_rpv_report.pdf (accessed 10 October 2026).

Seoul Metropolitan Government and KT (2026) *Seoul Living Mobility: Administrative-dong data*. Seoul Open Data Plaza [in Korean]. Available at: https://data.seoul.go.kr/dataVisual/seoul/seoulLivingMigration.do (accessed 10 October 2026).

Taiwan (n.d.a) 公職人員選舉罷免法 [Civil Servants Election and Recall Act], Article 17. Available at: https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=D0020010 (accessed 10 October 2026).
Taiwan (n.d.b) 總統副總統選舉罷免法 [Presidential and Vice Presidential Election and Recall Act]. Available at: https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=D0020053 (accessed 10 October 2026).